

Arpan Saha

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EDUCATION

McGill University | Montreal, QC, Canada

September 2021 - August 2025

B.A. in Computer Science with minor in Statistics

- **Leadership Roles:** VP IT, SRI McGill; Volunteer, Care Through Tech; Captain, intramural soccer and volleyball
- **Relevant Courses:** Database Systems, Software Design, Data Science, Machine Learning, NLP

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C++, R, HTML/CSS, SQL

Frameworks & Tools: React, Next.js, Flask, Spring boot, MySQL, Docker, AWS (EC2, S3), Bash, Git, GitHub Actions

Libraries: Tailwind CSS, NumPy, Pandas, PyTorch, TensorFlow, Scikit-learn, Selenium, Matplotlib, JUnit

Languages: English, French, Bangla, Hindi

EXPERIENCE

Web Developer

Apr 2022 - Apr 2025

McGill University HR | Montreal, QC, Canada

- Revamped **2,000+ webpages** in **Drupal** CMS by applying **WCAG** accessibility standards and responsive design best practices, resulting in significantly improved usability and compliance.
- Led a major site-wide revamp, boosting engagement by **350%** (15k to 65k monthly visitors) through improved UI design, content architecture, navigation and SEO.
- Built real-time dashboards in **Google Analytics** to monitor **SEO**, optimizing for **Google PageSpeed** scores.
- Developed a **Selenium**-based tool to automatically detect accessibility issues and recommend improvements, reducing manual testing workload.
- Led cross-departmental projects to deliver scalable web solutions, strengthening communication effectiveness.

ML Research Assistant

Dec 2022 - April 2023

McGill University | Montreal, QC, Canada

- Developed **convolutional neural network** model to segment medical images in soft-tissue sarcoma research.
- Collaborated with Dr. James Tsui to develop and evaluate ML models to optimize radiotherapy treatment.

PROJECTS

CampusVibe <https://github.com/ArpanSaha07/campus-vibe>

- Built a campus events management web app with **Next.js + TypeScript + Tailwind CSS** frontend (deployed on **Vercel**) and **Java Spring Boot** backend (deployed on **AWS Elastic Beanstalk**).
- Implemented **Spring Security + OAuth**, managed database schemas with **PostgreSQL + Flyway** on **AWS RDS**.
- Integrated **AWS S3** for media storage; tested with **JUnit, Mockito**, and **Playwright**.
- **Dockerized** deployment and automated CI/CD pipelines with **GitHub Actions**.
- Leveraged **GitHub Copilot** and **ChatGPT** to accelerate development, resolve issues, and enhance code quality.

Siloing of Political Thought in Social Media github.com/ArpanSaha07/Siloing-of-Political-Thought-in-TikTok

- Authored a research paper investigating how political content silos form on TikTok by creating **Python** bots that mimicked human interactions to collect and analyze large datasets, leading to key insights on content clustering.

SignLingo <https://github.com/abrarfuad27/SignLingo>

- Developed interactive Sign Language learning app using **React, Flask, SQLite, Pandas, TensorFlow, Docker**.
- Implemented hand gesture recognition using CNNs, with real-time practice xy and feedback.

FITnFLEX <https://github.com/visnunathan8/FITnFLEX>

- Created a wellness app to track fitness, meditation, and dietary habits with real-time feedback.
- Integrated **Angular, Flask, MySQL, Google OAuth** and **OpenCV** to create a seamless user experience.

Chelf <https://github.com/smrahman1/Chelf>

- Built an inventory management app recommending recipes based on available ingredients.
- Technologies used: **Python, React, Flask, PostgreSQL, OpenAI API, Figma**. Developed with a team of four at McHacks Hackathon 2023.